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Spikes in UK wildfire emissions driven by peatland fires in dry years

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Abstract

Wildfires on peatlands can nearly double global fire-driven carbon emissions, requiring centuries to re-sequester carbon (C) losses. Peatland fires require sufficiently hot, dry conditions and/or drainage for the peat to burn. Although these conditions have historically been infrequent, the warming and drying climate could increase the potential for wildfires and subsequent emissions. Here, we evaluate how climate change impacts peatland fire emissions by using the United Kingdom as a case study—where peatlands store an estimated 3.2 PgC. We use a fire emission model to quantify fire-driven C emissions using high-resolution land-surface data and fire-weather indices. Between 2001 and 2021, we estimate 0.8 TgC has been emitted from fires on peatlands, which can contribute up to 90% of total annual UK fire-driven C emissions. Consequently, protecting peatlands from fires in the UK would be a cost-effective way to slow climate change by avoiding future emissions. Peatland emissions spike during prominent dry years, implicating the inter-annual climate as a dominant driver of year-to-year variability. Integrating future climate projections suggests that a 2 °C global warming level could increase fire-driven C emissions in peatlands by over 60% solely via increased burn depths. Our findings are likely a bellwether for other temperate peatlands where climate change is leading to drier conditions, which increase burn depths and C emissions.

1. Introduction

Human-induced climate change has increased the risk of wildfire occurrence through increasing the frequency and severity of fire-weather (Abatzoglou and Williams 2016, Jones *et al* 2020, van Wees *et al* 2021, Zhuang *et al* 2021). Consequently, ecosystems that rarely burned are now burning more frequently where climates have become drier (Walker *et al* 2019). When these ecosystems are carbon (C) dense, C emissions can spike (Poulter *et al* 2017). Peatlands store ~33% of the planet's soil carbon, whilst only covering ~4% of the world's land surface (UNEP 2022). Yet, much of our fire-driven peatland C emission estimates rely on parameters from tropical peatlands, where drained peatlands are exposed to hot, dry conditions and experience burn depths two-fold greater than at temperate sites. For example, UK peatland fires have burned to an average measured depth of ~8 cm in three of the UK's largest peatland wildfires (e.g. see Davies *et al* 2013, Belcher *et al* 2021, Cui 2022), compared to ~33 cm in Indonesia (Ballhorn *et al* 2009). Refining the impact of fire on peatland C within temperate regions is key given the increasing temperatures and likelihood of extreme soil-moisture drought (Caretta *et al* 2022) facilitating wildfires on peatlands to become more severe.

Here, we used the UK—constituting ~13% of the world's blanket bog—as a testbed for evaluating the trends in fire-driven C emissions in temperate peatlands, and the role of climate change in shaping future emissions. Peatlands cover 9% of the UK's land-area (Xu *et al* 2018), sequestering >3 Tg CO_2 per year when in a 'healthy condition' (IUCN UK Peatland

Programme 2009). In the UK, climate change is projected to increase the risk of wildfires (Gazzard *et al* 2016, Jones *et al* 2020), which have already occurred on peatlands (e.g. the 2018 Saddleworth Moor fire and the 2019 Flow Country fire). If wildfires continue to increase and degrade peatlands, wildfires could reduce an important C sink. Consequently, a better understanding is needed of the drivers of wildfire emissions, the role of peatland fires, and how these may change in the future.

In non-peatland ecosystems, aboveground vegetation composition and fuel load play an important role in determining fire severity, and aboveground C emissions are fairly predictable. In peatlands, combustion of soil organic carbon (SOC) can be an important source of C emissions, but depends on burn depth, and can vary from ∼1–5 $kgCm^{-2}$ within pristine peatlands, to ∼25 $kgCm^{-2}$ if drained (Wilkinson *et al* 2023). Burn depth is largely driven by peat moisture and thus dependent on climate (Macrae *et al* 2012) compounded by local factors such as topography and management (Wilkinson *et al* 2023).

In wetter climates, such as the UK, sufficiently dry weather conditions are important for ignition and spread—particularly important in peatlands where soil moisture is generally saturated (Millard *et al* 2018). In many regions of the world, anthropogenic climate change has led to increases in fire-weather indices, and fire-weather seasons have lengthened (Jolly *et al* 2015, Jones *et al* 2020). Whilst no significant long-term trends in fire-weather and burned areas have been identified for the UK, summer months of very high fire danger were found to be closely related to burned areas for the years 2003–2020 (Perry *et al* 2022). Furthermore, spring fires were found to have a moderate positive relationship with indices, such as the Initial Spread Index (ISI), which integrates fine-fuel moisture and wind (Perry *et al* 2022). Linking these metrics to C emissions, and partitioning these emissions between aboveground and SOC may provide an avenue for better predicting fire-driven C emissions in peatlands.

Here, we produce the first bottom-up estimate of fire-driven C losses in the UK from 2001–2021. Fire-activity data show peatland fires comprised ∼25% of total average burned area from 2001–2021, occurring in every nation of the UK. To estimate the C emissions, we combined high-resolution data on peatland extent (defined as peats deeper than 0.5 m), vegetation type and agricultural land cover (Morton *et al* 2011, 2020a, 2020b, 2020c, 2020d, 2020e, 2021, 2022, Rowland *et al* 2017a, 2017b), soil moisture and fire occurrence, with an emission model parameterized for UK ecosystems. We then test the hypothesis that fires, especially in peatlands, positively correlate with inter-annual variability of dry climate conditions.

2. Methods

2.1. Calculating carbon emissions

Emission calculations commonly comprise spatially explicit vegetation cover maps combined with burned area estimates taken from satellites. These are then coupled with estimates of fuel load and fuel consumption within burned areas, matching fire data with climate data (such as soil moisture) to calculate emissions (e.g. van der Werf *et al* 2017, van Wees *et al* 2022). Here, the Global Fire Emissions Database (GFED) provides a benchmark for emissions, used as a validation tool for other estimates and in Intergovernmental Panel on Climate Change reports (van Wees *et al* 2022).

In this study, we calculated UK fire emissions using high-resolution data on land cover and burned area, combined with a static aboveground carbon map of Great Britain and Northern Ireland to provide a bottom-up estimate of C emissions from fires between 2001 and 2021. As in GFED and van Wees *et al* (2022), to calculate emissions, fuel loads are multiplied by satellite-derived burned area and a metric for combustion completeness (CC) (Seiler and Crutzen 1980).

To map the fires, we use a combination of the FireCCI51: MODIS Fire_cci Burned Area Pixel Product, Version 5.1 (Parellada 2018) and the European Forest Fire Information System (EFFIS) to provide monthly burned area at 250 m resolution. Both of these products provide a high ∼250 m resolution burned area and have been shown to be in the best agreement with each other in terms of burned area estimates (Turco *et al* 2019).

To calculate emissions for aboveground fuel loads, we utilize published model estimates of above-ground carbon for Great Britain taken from Henrys *et al* (2016). In Henrys *et al* (2016) the carbon within the different vegetation types is calculated by estimating the carbon density for each Great Britain land cover type, where the carbon density for wooded areas is age and species specific. This therefore provides a high-resolution carbon density map specific to the above-ground vegetation of Great Britain. Since this does not yet cover Northern Ireland, a reference above-ground biomass map for 2010 from Spawn *et al* (2020) was used. The Spawn *et al* (2020) data set provides a reference biomass for which modelled biomass is compared against in van Wees *et al*'s (2022) 500 m spatial resolution emission calculations, and provides integrated above- and belowground biomass maps that encompass all vegetation types (van Wees *et al* 2022: 12). Following GFED (van der Werf *et al* 2010), the amount of aboveground carbon within each burned area grid cell is apportioned between tree and non-tree vegetation; we do not account for emissions from the burning of surface litter fuels.

We mapped UK-specific vegetation types using a series of land cover maps (taken in 2007, 2015, 2017, 2018, 2019 and 2020) and calculated the fraction within each of the burned areas. This enabled increased accuracy of the fuel available to burn, whilst also removing areas that present as lakes, bare ground and urban fires. To capture vegetation changes over the studied period of 2001–2021, a series of high-resolution (10–25 m spatial resolution) vegetation maps were used: 2007 (Morton *et al* 2011, 2014), 2015 (Rowland *et al* 2017a, 2017b), 2017 (Morton *et al* 2020a, 2020b), 2018 (Morton *et al* 2020c, 2020d), 2019 (Morton *et al* 2020e, 2020f), 2020 (Morton *et al* 2021, 2022). The fraction of each vegetation type was calculated within each burned grid cell, and then aggregated to provide four main land cover categories comprising 'Forests and Woodlands', 'Moorlands and Heathlands', 'Peatlands' and 'Other Natural and Managed Lands', and aboveground carbon apportioned appropriately (figure S1). Here, for example, the land cover 'semi-natural grassland', 'heather grassland' and 'heather' were grouped into 'moorland and heathlands'; 'improved grassland' and 'arable' were grouped into 'other natural and managed lands' whilst 'conifer' and 'mixed and broadleaf' were grouped into 'forests and woodland' and 'bog' (which is categorized as peat >0.5 m deep by the respective land cover maps) into 'peatland'.

From this, we were able to calculate the proportion of land cover types within each burned area grid cell. The fraction of each aboveground carbon 'pool' combusted within each grid cell is denoted by the CC (van der Werf *et al* 2010). Here, we used the maximum and minimum CC values determined by van der Werf *et al* (2010) for the different fuel types to provide a 'maximum' and 'minimum' emission scenario for the UK (figure S1).

The carbon density for each burned area grid cell was then extracted from the centroid. The carbon density of the aboveground vegetation (given as t/ha) was then apportioned depending on the calculated fraction of the vegetation type within each burned area grid cell where, following GFED (van der Werf *et al* 2010), the amount of aboveground carbon within each burned area grid cell is first apportioned between tree and non-tree vegetation.

Fire C emissions are then calculated for each vegetation class within each burned pixel, by multiplying the burned fraction and the available carbon with the CC and the fraction of carbon in fuel types (see SI for further details). We evaluated the sensitivity by comparing maximum and minimum CC. To calculate SOC emissions from burning in peatlands, we used the same equation as that used to calculate SOC burned at 500 m resolution in van Wees *et al* (2022), whereby the amount of SOC burned per pixel is equal to the burn depth multiplied by the peat C-bulk density and the fraction of peat in a burned grid cell (we evaluate the sensitivity of bulk densities by comparing a range of values).

Peat burn depths were estimated using a similar linear function to van Wees *et al* (2022), whereby burn depth is dependent on the soil moisture content (here we used the ERA5 land monthly average volumetric soil water (Muñoz Sabater 2019)) tied to published in-field average burn depth measurements (equation (1)):

$$\text{SOC burn depth (cm)} = 13.88 * \text{soil dryness}_{0-7\text{cm}} - 3.024. \quad (1)$$

Using this equation, at minimum moisture conditions burn depths reach ~9.0 cm maximum (figure S2). Although this burn depth has been recorded in the natural peatlands of the UK Flow Country wildfire (Cui 2022), we limit the burn depth to 8.0 cm to provide a conservative emission estimate. This also minimizes any C emissions from smaller management burns conducted in wetter months (commonly October–March) over land cover defined as 'peatland', which does not burn into underlying peat layers (following the Heather and Grass Burning Codes (DEFRA, 2007, Welsh Assembly Government 2008) and the Muirburn Code (SEERAD 2008) (Worrall *et al* 2010). We then use the same equation as that used in van Wees *et al* (2022), whereby the amount of SOC burned per 250 x 250 m grid cell and therefore the amount of carbon lost/emitted (Mg C) is equal to the burn depth (m) multiplied by the peat carbon bulk density (Mg m^{-3}), the fraction of peat within the grid cell (m^2) and a CC (equation (2)):

$$\text{Carbon emitted from SOC burning} = P * \text{CBD} * \text{peatland fraction burned} * \text{CC}. \quad (2)$$

Equation (2): P = depth of burn based on soil moisture scalar, CBD = peat carbon bulk density (nation specific using nation-specific peat bulk density and carbon content (see table S1)), CC = combustion completeness, for which we use 0.5.

CC of 1.0 is commonly adopted when calculating C emissions from peatland fires, particularly in the tropics (e.g. Poulter *et al* 2006). However, it is often acknowledged in the literature as an oversimplification (Krisnawati *et al* 2021, Volkova *et al* 2021). Peatlands can exhibit significant microtopographic variation with dry, high hummocks and low, wet hollows and as a result, have been suggested to vary in their vulnerability to combustion and respective carbon losses during combustion (Benscoter and Wieder 2003, Wilkinson *et al* 2019). For example, within western Canadian bog hummocks were found to experience almost twice as much organic matter loss compared to hollows from combustion (Benscoter and Wieder 2003). The Intergovernmental Panel on Climate Change (IPCC 2014) therefore recommends that country-specific CCs be applied to improve the SOC emission accuracy where possible.

Whilst there is little information on defining CC for peat soils, particularly within temperate sites, a recent study conducted by Krisnawati *et al* (2021) based on in-field measurements estimated that CC is more likely to range between 0.40 and 0.68. As such, for the purpose of providing a conservative estimate of UK peatland C emissions, where burn depths are a fraction of those experienced in tropical peatlands, a CC of 0.5 was adopted.

2.2. Fire weather index (FWI) and ISI

Both the FWI and the ISI were taken from the daily Copernicus ERA5 reanalysis FWI and ISI data, on a 0.25 degree spatial resolution and aggregated to provide monthly averages across the UK. The ERA5 FWI reanalysis data is based on the Canadian FWI system fire danger model, which consists of different components that account for variables that influence fire ignition and fire spread. Each of these variables, including fuel moisture, temperature, relative humidity, wind and rain effects on fire spread and behaviour, are used to provide an estimate of fire danger, where the more favourable the meteorological conditions to fire ignition and spread, the higher the FWI.

The ISI provides a 'rating of expected rate of fire spread' (Perry *et al* 2022:562) and is reflective of the fine-fuel surface vegetation moisture content, combining the effects of wind as well as fine-fuel moisture.

2.3. Projected soil moisture changes at 2 °C global warming level (GWL)

Projections of future soil moisture were used from Kay *et al* (2022) where 1 km grids of monthly mean soil moisture were simulated, driven by the UK Climate Projections 2018 regional projections (see Kay *et al* (2022) for more information). Here, the mean over 20 years was taken, centred on the crossing time of a GWL of 2.0 °C (Arias *et al* 2021) and under each of the different ensemble members used in (Kay *et al* 2022) (Ensemble members 1; 4, 5, 6, 7, 8, 9, 10, 11, 12, 13 and 15). Burned area was not altered, so that only soil moisture and the resulting SOC burn depth and resulting C emissions were calculated. The UKCP18 Regional Projections were used to drive a hydrological model to produce soil moisture projections (Kay *et al* 2022), using climate projections comprising a 12-member perturbed parameter ensemble of the Hadley Centre RCM for December 1980–November 2080 forced by the RCP8.5 emission scenario. Each member is produced from varying parameters allowing for uncertainty in future projections from climate modelling to be explored (Arias *et al* 2021). Examples include perturbations in cloud physics, convection scheme, surface processes, radiative transfer, etc, with ensemble 01 representing the standard parameterization. Each member has the potential to influence projected rainfall patterns in different ways across the UK, with some areas experiencing earlier drying dates (increasing the amount of time in a year with drier soils), and a general trend of lower precipitation in the fall, with the exception of western Scotland that is projected to experience an increase in precipitation (see figure 6 in Kay *et al* 2022).

3. Results

3.1. UK annual fire carbon emissions varied significantly over 2001-2021

Overall, we estimate fires emitted 44.8 kilotonnes of C (ktC) annually on average between 2001 and 2021 (figure 1(a)). Unsurprisingly, there is a strong relationship between C emissions and total burned area across the UK (Pearson's $R = 0.95$, $p < 0.0001$). Inter-annual variability was high, producing sporadic spikes in emissions. For example, emissions in 2003 totalled 337 ktC, nearly eightfold higher than the mean. Years 2007, 2009 and 2019 also had spikes in emissions. For total burned area, 2003 and 2019 had the largest total burned areas of 50 993 and 29 396 ha, respectively. To understand the variability in emissions, we evaluated whether it was due to variations in total burned area, the ecosystems that burned, and/or the climate conditions that month.

Climate is one important driver of fire activity. We found that total burned area in the summer-fall months (June–November) was positively correlated with the summer-fall monthly averaged FWI (figure 1(b); $R^2 = 0.4$, $p < 0.01$). The FWI approximates how favourable the meteorological conditions are for fire ignition and spread (Perry *et al* 2022). In addition, we found that the burned area in winter-spring (December–May) was correlated with the monthly averaged ISI. The ISI quantifies the expected fire spread rate, considering wind and fine-fuel moisture (figure 1(c); $R^2 = 0.2$, $p = 0.05$). Thus, favourable meteorological conditions for fire ignition and spread in summer (FWI) combined with more favourable conditions for fire spread (ISI) positively correlate with total burned area. We next sought to evaluate the contribution of different ecosystem C pools to the emissions to further unpack the mechanisms contributing to emission variability.

3.2. Carbon emissions from burning into organic soils dominates trends in UK fire emissions

On average, the C emissions produced from combusting SOC comprise ∼70% of the total UK C emissions from fires between 2001 and 2021, with the contribution even reaching 81% in one year (figure 2(a)). The variability is due to both the soil moisture conditions suitable for SOC combustion combined with the ecosystem being burned and its underlying SOC stocks. Estimated SOC burn depths ranged between 0.9 and 8.0 cm between the highest and lowest soil moisture values, respectively. Thus, while climate may drive much of the inter-annual burned area and absolute

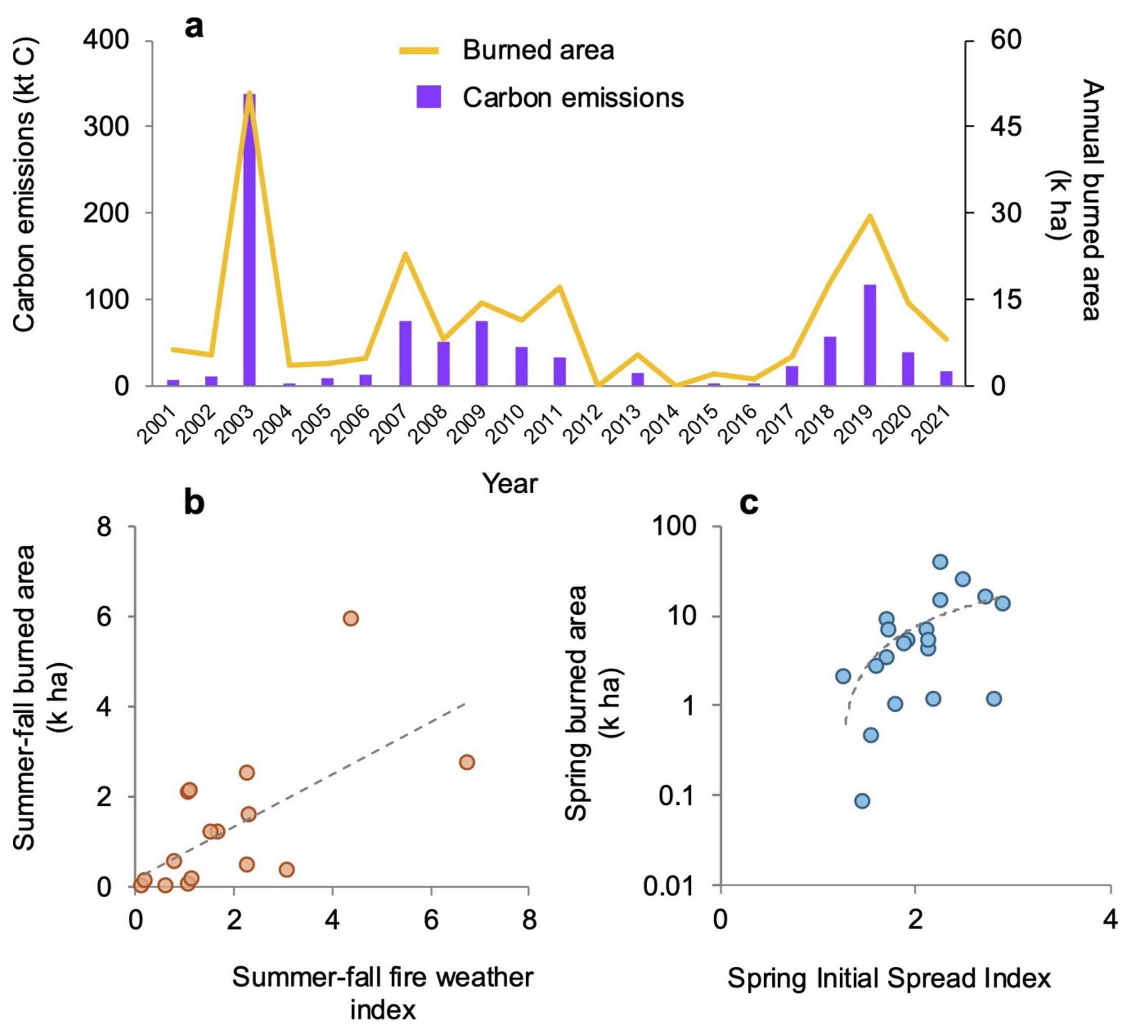

Figure 1. (a) Total annual fire-driven carbon emission (kilo tonnes of carbon) for the United Kingdom (under maximum combustion completeness) (bars), and variability in burned area (thousands of hectares) for the UK between 2001 and 2021 (line). (b). Relationship between summer-fall burned area and mean summer-fall Fire Weather Index $R^2 = 0.4$; $F_{1,14} = 9.3$, $p < 0.01$. (c). Relationship between yearly average spring Initial Spread Index (ISI) and spring burned area; note y-axis is on a log scale and statistics conducted on log-transformed data, $R^2 = 0.2$ and $P = 0.04$. Each point represents a year in panels b and c.

emission variability, the source of the emissions—SOC combustion—is relatively consistent (with the exception of 2014 when no fires were recorded over a peatland land cover) (figure 2(b)).

The ecosystem that burned also contributed to trends in C emissions. When considering the vegetation type within each burned area grid cell (figure 2(b)), we found that moorlands and heathlands contributed the most to the total burned area in 18 out of the 20 study years, averaging 5600 ha per year (95% CI of the sample mean, 2886–8346 ha). Other natural and managed lands comprised 1082 ha (95% CI of the sample mean, 558–1607 ha) and forests and woodlands comprised 745 ha (95% CI of the sample mean, 415–1076 ha) per year. Peatlands burned on average 2530 ha per year (95% CI of the sample mean, 684–4376 ha). Thus, moorlands and heathlands dominated burned area—likely due to their large spatial extent and flammability, combined with managed fire frequencies of burning every ∼10–25 years—the contribution of intentional burns is likely mixed since the fires detected from satellites are relatively large and management burns can be small (Thacker *et al* 2015). Despite dominating the burned area, moorlands and heathlands were not always the dominant contributor to total fire-driven C emissions because of the contrasting patterns in aboveground C stocks versus SOC-dominated C emissions.

Emissions on moorlands and heathlands were primarily due to aboveground biomass combustion, with aboveground biomass pools having high CC (80%–100%) (van der Werf *et al* 2010, 2019), yet low C density. When only considering aboveground C emissions, moorlands and heathlands dominated in 13 out of the 20 years studied, with mean emissions of 3.1 ktC (95% CI, 2.0–5.8 ktC) (figure 2(b)) under minimum CC conditions, and a mean of 3.9 ktC (95% CI, 1.6–4.6 ktC) under maximum CC conditions. Importantly for the C budget, these aboveground C losses are short term and can be sequestered once vegetation re-grows postfire (Legg *et al* 2010). Only in 2006 and 2009 did forests and woodlands

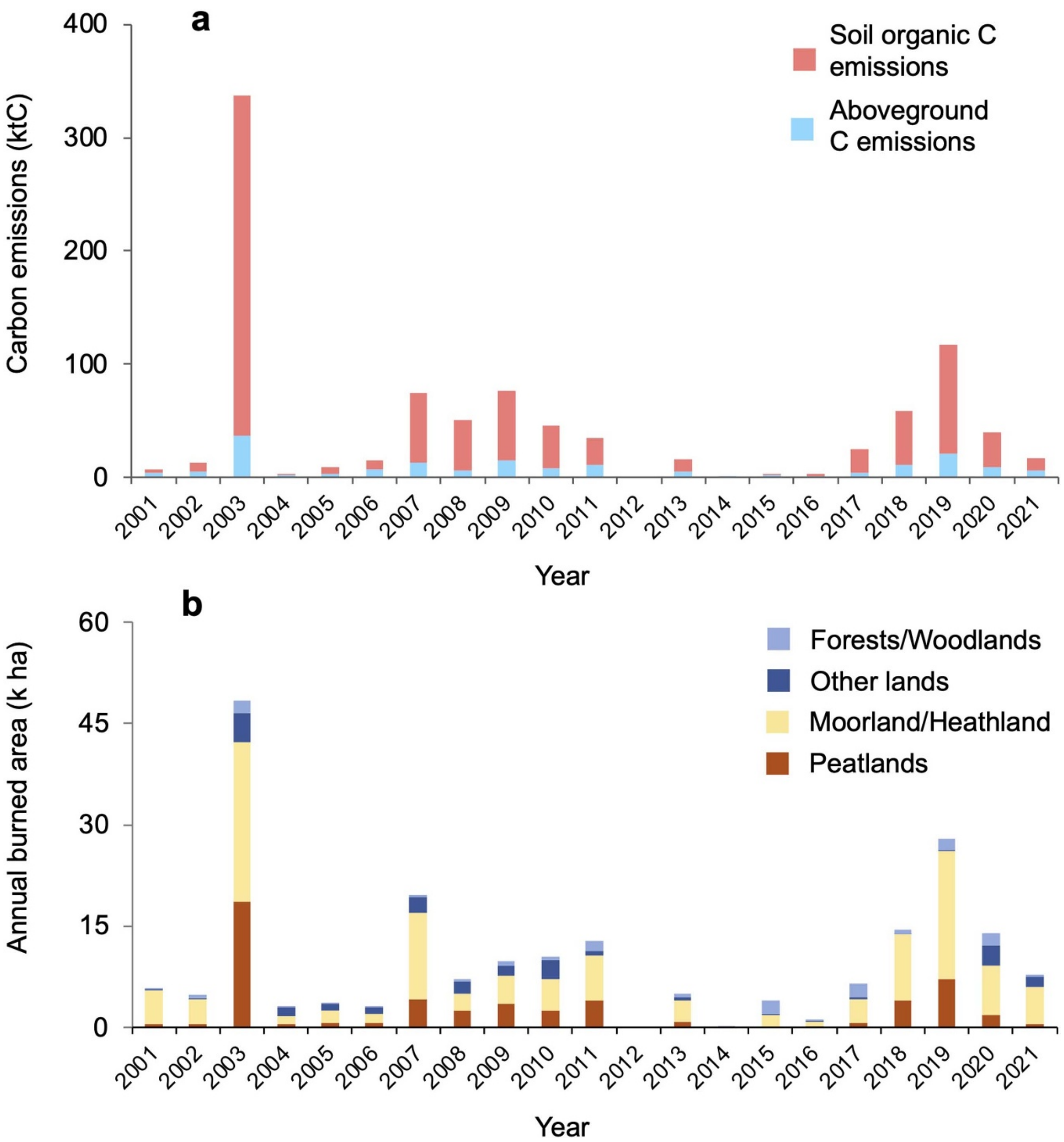

Figure 2. Carbon emissions are highly dependent on soil organic carbon losses and burned area in peatlands. (a) Cumulative fire-driven carbon (C) emissions for the United Kingdom separated into belowground and aboveground emission sources. SOC = soil organic carbon (b) Cumulative annual area (in thousands of hectares) of available fuel to burn summed across all burned grid cells.

contribute more than moorlands and heathlands to aboveground C emissions (5.1 and 7.0 ktC, respectively) (figure 2(b))—likely due to the lack of burned area in moorlands and heathlands.

However, when SOC emissions are considered, peatlands are the dominant source. In fact, in years where the aboveground emissions peaked—2003, 2007, 2009 and 2019—SOC emissions also peaked and exceeded the aboveground emissions. Specifically, the aboveground emission estimates reached 29.2–36.6 ktC in 2003 and 16.5–20.6 ktC in 2019, whilst SOC emissions reached an estimated 300.8 ktC in 2003 and 96.5 ktC in 2019. Hence, when considering total fire-driven C emissions, peatlands are an important contributor to inter-annual variability. Peatland emissions are primarily comprised of SOC combustion, which has a high carbon density despite variable burn depths (ranging from 0.9–8.0 cm). Thus, fires on peatlands emit more C than fires in the surface fuels of moorlands and heathlands in dry years.

Since C emissions from peatlands depend on burn depth, which is related to soil moisture, we conducted sensitivity tests to refine our estimates of SOC losses by varying the C-bulk density and the relationship between soil moisture and the equation for maximum burn depth in soils (figures 3(a) and (b)). Although these assumptions change the absolute magnitude, overall trends remain, so that when SOC emissions spike, so do total annual emissions.

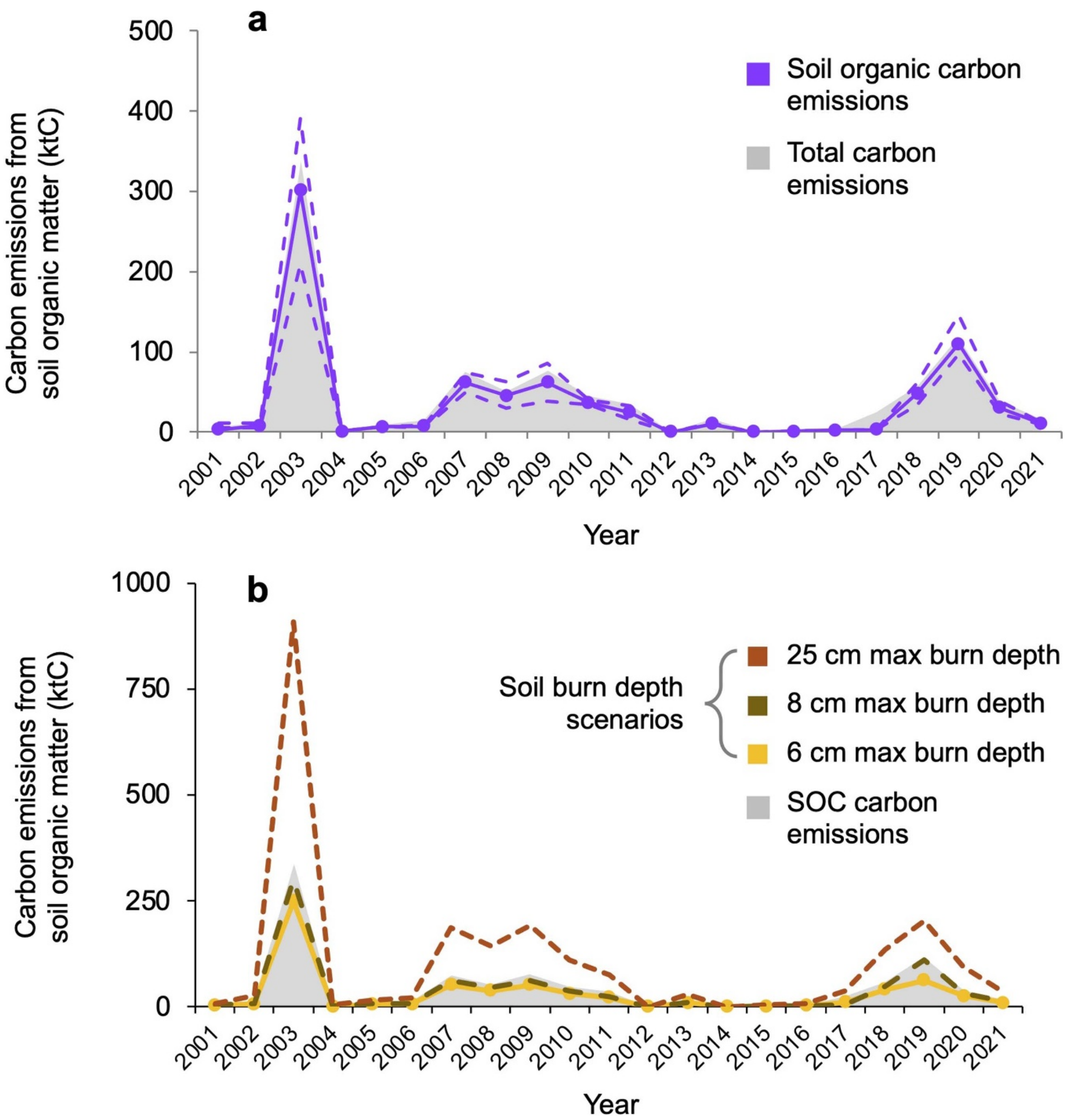

Figure 3. Sensitivity analyses for carbon (C) emissions from soil organic matter. (a) soil organic C (SOC) emissions from combustion of soil organic matter sensitivity to the variations in bulk density by re-calculating emissions using the standard errors around the mean bulk density (see section 2, table 1 for details) with the dashed line illustrating the ± standard error around the mean. (b) Effect of variations in the relationship between soil moisture and maximum burn depth. Equations are as follows: maximum burn depth of 6 cm (yellow line, burn depth = 10.87 * soil-dryness—2.15), maximum burn depth of 8.0 cm (brown line and points; equation (1) used in this study), and maximum burn depth of 25 cm (orange line; burn depth = 48.98 * soil dryness—13.2).

Given the importance of the sporadic spikes in SOC emissions for total fire-driven C emissions, we next sought to identify the conditions that determine fires on peatlands and their associated emissions. The FWI averaged within a year was positively correlated with total SOC emissions (figure 4) ($R^2 = 0.26$, $p = 0.01$). The year 2003 was exceptional, with both the highest estimated SOC emissions and the highest FWI out of the 20 years studied (+200% higher than the mean for 2001–2021).

3.3. The number of UK fires is increasing and the fire season is lengthening

The number of fires has increased year-on-year between 2014 and 2021, with the exception of a decline from the 137 recorded fires in 2019 to 114 in 2020, before more than doubling to 234 between 2020 and 2021 (figure 5). A Mann–Kendall test reveals that between 2014 and 2021 the trend is significant ($p = 0.002$), although the sample size is small, being only 8 years worth of data, due to the limited length of the EFFIS data.

Fire seasons are also lengthening, where the number of months recording burned area rises from 1 to 4 months between 2011 and 2016 to between 6 and 9 months between 2017 and 2021, with the years 2020 and 2021 experiencing the longest fire season (figure 5). In general, 69% of the fires occur between March and May. From 2016, fires were also recorded during the summer months (June, July and August)

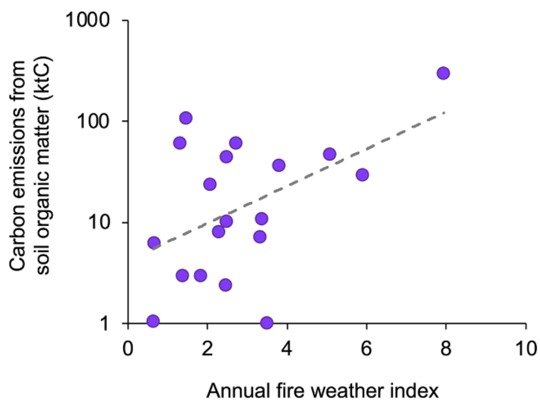

Figure 4. Relationship between total soil organic C emissions and average annual Fire Weather Index. Note y-axis is on a log scale and linear model fit on log-transformed data. $R^2 = 0.26$, $F_{1,14} = 8.4$, $p < 0.01$. Each point is a year.

and into the fall months (September, October and November) from 2017. The changes differed across the nations, where the emergence of a new fire season was the largest in Scotland (where peatlands comprise ~23% of land-area (Novo *et al* 2021)) but also present in Wales and England. Given that Scotland comprises ~45% of the total UK burned area, the lengthening of the fire season is important.

3.4. Calculating SOC emissions under future soil moisture projections

It is projected that the UK will experience a greater chance of warmer and wetter winters combined with hotter and drier summers (Lowe *et al* 2018). The implications for fire are high, with the number of summer days experiencing 'very high' fire danger is projected to double under a 2 °C GWL scenario (Perry *et al* 2022). Using a UK grid-based simulation of soil moisture (Kay *et al* 2022), and applying our SOC burn depth regression, we calculated C emissions if fires occurred in the same peatland areas that we observed burning historically using a GWL of 2 °C, taking a 20 year mean of simulated soil moisture, under different climate ensemble members (see section 2) (figure 6).

All ensemble members produced similar projections, where annual emissions were, for most years, higher than current SOC-emission estimates (figures 6(b) and (c)). Total C emissions (for burned area equivalent of 2001–2021, but under future climate conditions) ranged from 1031–1163 ktC between the different ensemble members tested. Taken across all members, future climate conditions under a 2 °C global warming scenario are projected to increase the UK's C emission by 61% compared with 2001–2021 emissions, with a median of 18.3 ktC projected compared to a median of 11.1 ktC under historical 2001–2021 emissions, even without changing the total burned area or the locations where fires occurred.

4. Discussion

We found that fire-driven C emissions were explained by two processes: (i) exceptional spikes in emissions were driven by greater burn depths in peatlands, (ii) higher overall emissions were associated with greater burned area, partly due to lengthening fire seasons, combined with higher combustibility of aboveground plant biomass. Both of these processes were associated with drier weather conditions. The sensitivities of emissions to assumptions around weather, soil moisture and burn depths highlights a critical research gap that needs addressing. While other biomes may dominate total burned areas, such as moorlands, and potential rising baselines in annual emissions, their C emissions can be vastly superseded by deep burns on peatlands.

Spikes in annual C emissions were largely attributable to years of high SOC combustion, occurring under dry weather conditions. Specifically, we find that SOC emissions have significant positive correlations with increasing yearly averaged FWI, as well as a positive correlation with spring ISI, where meteorological conditions, including fuel moisture, are more favourable for fire ignition and spread. One of the best examples of burned area associated with dry weather conditions is 2003, where the highest annual emissions were associated with high FWI within burned areas (197% higher than the mean for 2001–2021). 2019 recorded the second highest burned area and emissions, yet the fire-weather across the entire year was not exceptionally high. However, the 2019 fires were mainly in the spring (particularly April) and whilst neither the FWI nor ISI was particularly high for the UK as a whole, the ISI was high for Scotland (averaging 3.4 and maximum ISI of 4.8 for April), which accounted for ~72% of the total burned area for this year. The importance of fire-weather conditions is also evidenced by the low emission years of 2012 and 2014, which coincide with particularly wet years and which our analysis suggests experienced no SOC-related emissions. These years experienced low C emissions from aboveground sources, with no fires recorded by EFFIS in 2012, and ~24 tC emitted in 2014. Historical records demonstrate 2012 as being the wettest summer since 1912 with record rainfall between April and July. 2014, whilst at the time being one of the warmest years on record, was also the UK's fifth wettest years since 1910. Thus, the timing of dry weather in the year combined with the location of the dry weather both play important roles.

In comparison, the springs of 2003, 2007, 2009 and 2019 were characterized by warm, dry conditions (Perry *et al* 2022). According to the Met Office monthly climate summary, the spring of 2009 was 1.2 °C above the 1971–2000 average, with below average rainfall over most of England and Wales. August 2003 also experienced a record-breaking heatwave (Luterbacher *et al* 2004). Each of these years presents

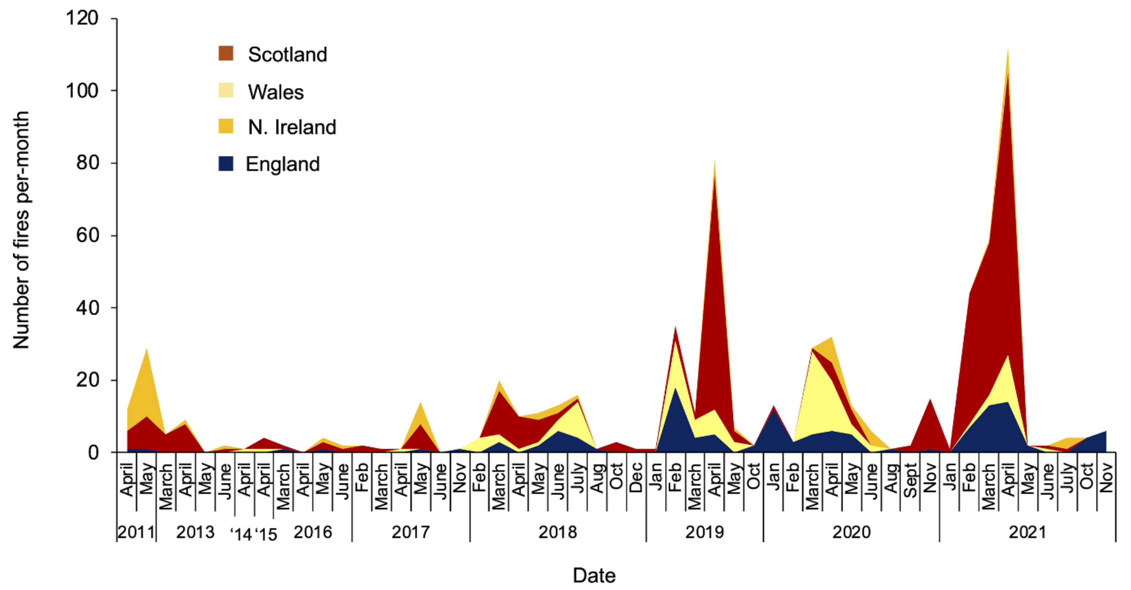

Figure 5. Number of fires occurring per month, per year and per nation between 2011 and 2021. There are no fires recorded by EFFIS for 2012. Note that the number of fires analyzed only covers 2011–2021 because although MODIS record the estimated day of the first detection of the fire for each pixel, determining how many days and distance between pixels qualifies as a 'new fire' is subjective, hence we do not quantify the MODIS data into number of fires.

as high fire emission years. Hence, it appears that dry years may be capable of shifting emission sources from aboveground to belowground.

Global fire activity is controlled by a number of factors including climate, ignition, biomass production (fuel) and its availability to burn (Bedia *et al* 2015). Climate change is capable of altering fire dynamics through an extension of the fire season and/or creating drier conditions, increasing the probability of available fuel to burn in areas that historically were too wet to burn (Bedia *et al* 2015, Canadell *et al* 2021). This impacts burned areas by allowing more areas to burn in any given year. For example, an annual increase in forest area burned has already been identified in Australia over the past 32 years, linked with more dangerous fire-weather conditions (Canadell *et al* 2021). This, coupled with an understanding that drier conditions can also lead to a lowering of the water table in peatlands and increase the extent, frequency and burn depth of peat fires (Turetsky *et al* 2014) highlights the sensitive nature of future C emissions from fires under anthropogenic climate change.

Emissions from belowground sources are complex. For example, peat burn depths and their resulting emissions are sensitive to their inorganic content; bulk density (Hu *et al* 2020), soil temperature, depth of water-table, total peat depth (e.g. see Wilkinson *et al* 2020), and overlying vegetation cover that can impact C stocks as well as C loss from burning (e.g. Kettridge *et al* 2015). Whilst we try to incorporate bulk density changes and account for peat depth by only analyzing peatlands >0.5 m deep, we acknowledge that our emission estimates for peatland burning cannot encompass this complexity. When comparing our estimates between case studies, we find that our peatland C emissions are often conservative. For example, in 2019 a wildfire in Scotland, the Flow Country fire in May, reportedly burned ~53.8 km^2 of peatland with an averaged measured burn depth of ~8 cm in natural peatlands, and 25 cm in the drained peatlands (Cui 2022). Our estimates suggest a burn depth of 1.8–2.0 cm based on ERA5_reanalysis soil moisture for May 2019 at this location, indicating that our estimate of 96 ktC for the year 2019 is highly conservative. Incorporating an 8 cm burn depth, which is still conservative given that some areas reached ~25–30 cm (Cui 2022), our emission estimates increase to ~137 ktC for 2019, with ~96 ktC attributed to the Flow Country fire alone. This is based on a calculated fractional area of 35.16 km^2 of peatland >0.5 m depth being burned (from the LC 2019 maps). Previous estimates produced for the Worldwide Fund for Nature (Wiltshire *et al* 2019) suggested a low-range to mean estimate of ~174 to ~290 ktC lost, assuming 20% of the area had been drained. If we assume the same, our estimates suggest ~195 ktC could have been emitted from this one fire. Our estimates therefore appear consistent with Wiltshire *et al* (2019). However, this highlights the sensitivity of C-emission estimates based on peatland conditions and depth of burning.

Our emission estimates for another prominent peatland fire occurring in 2018, the Saddleworth Moor fire, also appear to be on the conservative side, with our estimate of ~24 ktC released from SOC compared with estimates of ~37 ktC (Belcher *et al* 2021). Whilst conservative, our estimates highlight

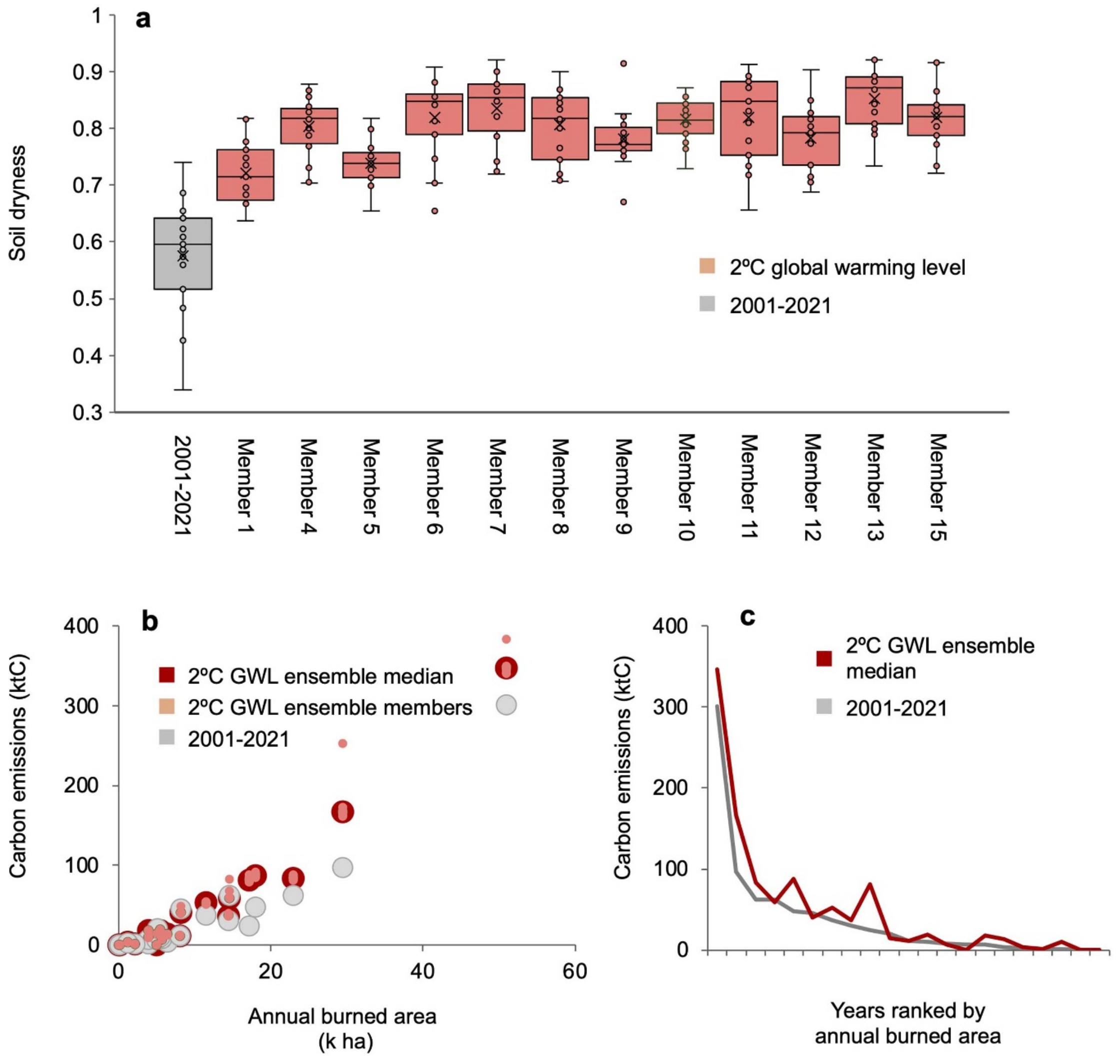

Figure 6. Yearly SOC emissions under future soil moisture projections (assuming no change in burned area). (a) Box plot illustrating changes in soil moisture between historical (taken between 2001 and 2021) and future climate conditions driven by 12 members of a climate model ensemble, under a 2 °C warming scenario using the same year-by-year burned area locations. (b) Scatter plot between burned area and the calculated C emissions under historical (2001–2021) climate conditions and future 2 °C global warming climate conditions represented as each ensemble member and the median of calculations, under a 2 °C global warming scenario using the same year-by-year burned area locations and amounts as observed between 2001 and 2021. (c) Ranked comparison between the 2001–2021 conditions and 2 °C global warming level ensemble median within a particular burned area distribution of a year. In other words, a within-scenario effect of 2 °C global warming level climate change on C emissions. 'Wetter' ensemble member projections produce slightly lower emission estimates in some years compared to 'drier' member projections, which produce higher emission estimates. For further details on the ensemble perturbations refer to Murphy *et al* (2018). More technical detail can also be found in Robinson *et al* (2023) and Lane *et al* (2024), which simulate projections of evaporation and soil moisture for the UK, respectively.

the importance of peatlands and their role in dominating UK fire C emissions, further emphasizing the importance of high-resolution emission estimates for areas, such as the UK, where peatlands make up such a large proportion. This emphasizes the need for fire management to remain an important consideration in managing peatlands to achieve net zero goals.

Our findings also highlight that there can be a decoupling between total burned area and total emissions. For example, moorlands and heathlands contributed the most to total burned area in 18 out of the 20 study years, averaging 5600 ha per year, but the years with the highest emissions were when SOC losses were the highest. Moorland and heathland burning is a contentious issue in the UK, with one argument against burning being the emissions from prescribed fires. However, the dominant emission source in these ecosystems is above ground in plant biomass, which tends to regrow in between burns, making it unlikely that they are a net source of emissions (note that our estimates do not account for the carbon drawn down by re-growth). Moreover, fire emissions are likely also buffered by the production of pyrogenic carbon, which can be buried for centuries to millennia in both terrestrial and marine pools (Jones *et al* 2019). One path forward would be to look to improve the quantification of the loss of C from these ecosystems that historically were too wet to burn, in order to be able to fully assess the impact of climate change on C emissions, particularly in areas that are most at risk of future wildfires.

5. Conclusion

With soil moisture consistent with a 2 °C GWL, we project that SOC emissions from fires on UK peatlands would increase by 61% compared to the current 2001–2021 data set. This is based only on projected drier soil conditions and assumes no change in burned area, which would undoubtedly push C emissions up further. In context, these emissions are equivalent to an average of ∼3.8 million tons of CO_2 projected to be released from peatland fires over the 20-year period, which is equivalent to adding average emissions of ∼133 commercial aircraft, 820 800 passenger vehicles and 414 000 homes (Bressler 2021). This highlights the sensitive nature of the future of the UK's C emissions, particularly as this assumes no increase in burned area, which would undoubtedly push emissions up further. Moreover, within the UK, C-dense soil deposits are shallower than the >0.5 m classification used in the UK-CEH land-cover maps (e.g. the Pentland Hills, Scotland) (Davies *et al* 2008). These C-rich peaty soils, and shallow peats (<0.4 m), may be at increased risk of wildfire due to their vulnerability to loss of moisture (Davies *et al* 2008, Wilkinson *et al* 2020), and may therefore be capable of contributing to equally high C emissions in the future.

Fire regimes are changing, notably driven by climate change, land use and human influences (e.g. ignition and suppression) (Rogers *et al* 2020). This is crucial since changing fire regimes may be capable of tilting the balance where emissions become increasingly dominated by the combustion of SOC

While much research focuses on tropical peatlands, high-latitude peatlands in the Northern Hemisphere are also at risk since temperatures are rising faster compared to other areas across the globe (Ma *et al* 2022). Hence, the widespread nature of northern peatlands, combined with the increasing threat of these ecosystems burning, means that these sites are increasingly vulnerable to producing high emissions. Thus, assessing the future wildfire risks and C emissions in areas that have high peatland land cover under a changing climate is essential, particularly as direct C emissions and combustion from soils are often not considered (Pellegrini *et al* 2022).

Data availability statement

Each of the data sets used to calculate aboveground C emissions are publicly available.

All data that support the findings of this study are included within the article (and any supplementary files).

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Ethics statement

We have no competing interests to declare.

Author information

S J B and A F A P designed the study and S J B conducted the study and analysis. R B and M P provided fire weather and climate data and contributed to analyses. S J B wrote the manuscript with contributions from all authors.

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